

Low-Cost 5V Power Supply

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All embedded systems require very little power to operate. Most of them operate at a wide voltage range (2 to 5.5V) with minimal current, but typically a regulated constant voltage of 5 volts is required to avoid malfunctioning of the circuit. Normally, regulators like

chips. The circuit is low-cost and easy to design as it uses few components.

Circuit and working

As shown in Fig. 1, the circuit is built around regulator TL431 (IC1), which is from three-terminal shunt regulator family, and a few discrete components. TL431 is a well-known adjustable regulator, combining reference voltage and an open-collector amplifier as shown in Fig. 2. Its internal active-output circuitry provides a

can be set by adjusting the ratio of R2 and R3 by following the relationship:

$$V_{out} = V_{ref} \left(1 + \frac{R2}{R3} \right)$$

where V_{ref} is the internal reference voltage of IC1 (2.5V).

The values of resistors R1 and R2 are chosen high enough to ensure minimal power loss. At the same time, these are low enough to achieve regulation accuracy and stability.

When the reference voltage of IC1 reaches 2.5 volts, its cathode sinks, causing the output from BC337 to drop. The moment the voltage at reference pin falls below 2.5 volts, IC1 allows transistor T1 to charge capacitor C2 to 5 volts. The no-load current of this circuit is in micro-amperes, whereas for a standard regulator it is in milli-amperes. This proves that it can significantly reduce power losses and hence improve efficiency, making it suitable for battery-operated circuits.

Working of the circuit is simple. It provides constant 5V output from an unregulated 9-40V DC input. The input power can be from a DC adaptor or a 9V/12V/24V external battery.

Construction and testing

An actual-size, single-side PCB for the cost-effective 5V regulator for embedded systems is shown in Fig. 3 and its component layout in Fig. 4. Suitable connectors are provided on the PCB for input and output. Assembling the circuit on a PCB minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Enclose the assembled PCB in a suitable cabinet. Fix a two-pin connector for 9-40V input at the back side of the cabinet and another two-pin connector for 5V output at the front side.

Check whether the input supply to the circuit is correct at test point TP1 with respect to TP0. The reference voltage is critical for the circuit and can be checked at TP2. The final output can be verified at TP3. ●

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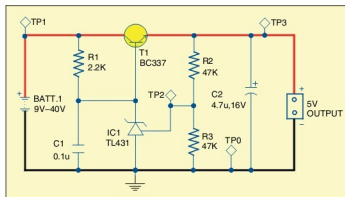


Fig. 1: Circuit of cost-effective 5V regulator

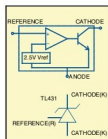


Fig. 2: TL431 IC configuration

7805 can do this job, but their quiescent current is high. Using this discrete circuit costing less than six rupees (except the PCB) is the best trade-off against the costly dedicated regulator

very sharp turn-on characteristic, making it an excellent replacement for zener diodes in many applications like on-board regulation, adjustable power supplies and switching power supplies.

The reference voltage at the reference pin of shunt regulator IC1 is set by the voltage divider network comprising resistors R2 and R3. An output voltage ranging between V_{ref} and 36V

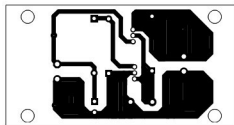


Fig. 3: An actual-size, single-side PCB for the cost effective 5V regulator

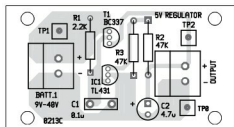


Fig. 4: Component layout for the PCB

Test Points

Test point	Details
TP0	GND
TP1	9-40V
TP2	2.5V
TP3	5V

PARTS LIST

Semiconductors:	
IC1	- TL431 shunt regulator
T1	- BC337 npn transistor
Resistors (all 1/4-watt, 45% carbon):	
R1	- 2.2-kilo-ohm
R2, R3	- 47-kilo-ohm
Capacitors:	
C1	- 0.1uF ceramic disk
C2	- 4.7uF, 16V electrolytic
Miscellaneous:	
	- 9V-40V variable power supply